

Eventual strict inequality in Horák’s rectangle conjecture

A partial result for the question in arXiv:2604.01138

11 August 2026

Status. candidate_partial_result_likely_valid_needs_human_review.

Abstract

For the Dirichlet p -Laplacian on $\mathcal{R}_a = (0, a) \times (0, 1)$, Horák conjectured a single transition from the second Ljusternik–Schnirelman eigenvalue to a lower nonsymmetric branch. For every fixed $1 < a < 2$, we prove the entire post-last-contact half: the second eigenvalue is strictly below the vertical antisymmetric branch for all sufficiently large p . We give an explicit sufficient inequality. The proof combines an opposite-corner two-disk packing, a genus-two family of distance cones, and sharp one-dimensional Poincaré inequality on the half-rectangle. The pre-contact equality interval and the regime $a \geq 2$ remain open.

1 Source conjecture and notation

For a bounded domain $\Omega \subset \mathbb{R}^2$, write

$$R_p[u; \Omega] = \frac{\int_{\Omega} |\nabla u|^p}{\int_{\Omega} |u|^p}$$

and let $\lambda_2(p; \Omega)$ be the second variational eigenvalue defined by the Krasnosel’skiĭ-genus minimax on the unit L^p -sphere. Following [1], let $\lambda_{\text{bar}}(p; \mathcal{R}_a)$ denote the branch whose eigenfunction is antisymmetric about $x = a/2$. By definition,

$$\lambda_{\text{bar}}(p; \mathcal{R}_a) = \lambda_1(p; \mathcal{D}_a), \quad \mathcal{D}_a = (0, a/2) \times (0, 1).$$

Remark 3.4 of [1], reproduced in Figure 1, records Horák’s conjecture: for every $a > 1$ there should be $p^*(a) > 2$ such that $\lambda_2 = \lambda_{\text{bar}}$ on $[2, p^*]$ and $\lambda_2 < \lambda_{\text{bar}}$ for $p > p^*$. The source anticipates $p^*(a) = +\infty$ for $a \geq 2$.

2 Proof intuition

Two tangent equal disks can be put into opposite corners of $\mathcal{R}_a$. Their distance-to-boundary cones span a two-dimensional, genus-two test family for λ_2 , and disjoint congruent supports make the Rayleigh quotient constant on that family. For $a < 2$, this tangent-packing radius is strictly larger than the inradius $a/4$ of the half-rectangle $\mathcal{D}_a$. That strict geometric gap becomes an exponential gap after taking p -th powers. A one-dimensional Poincaré bound on each horizontal slice of $\mathcal{D}_a$ then beats the two-cone upper bound for every sufficiently large p .

3 Explicit eventual inequality

Theorem 1. *Let $1 < a < 2$, and define*

$$r_a = \frac{a+1-\sqrt{2a}}{2}, \quad \rho_a = \frac{a}{4r_a}.$$

Then $0 < \rho_a < 1$, and for every $p > 1$ satisfying

$$\frac{(p+1)(p+2)}{2(p-1)} \rho_a^p < 1 \tag{1}$$

one has

$$\lambda_2(p; \mathcal{R}_a) < \lambda_{\text{bar}}(p; \mathcal{R}_a).$$

Consequently, the strict inequality holds for all sufficiently large p .

Proof. Place open disks B_1, B_2 of radius r_a at the opposite-corner centers

$$(r_a, r_a), \quad (a - r_a, 1 - r_a).$$

Since $1 < a < 2$, direct calculation gives

$$r_a < \frac{1}{2} < \frac{a}{2}, \quad r_a - \frac{a}{4} = \frac{(\sqrt{a} - \sqrt{2})^2}{4} > 0,$$

so both disks fit in $\mathcal{R}_a$. They are tangent because

$$(a - 2r_a)^2 + (1 - 2r_a)^2 = 4r_a^2.$$

For $i = 1, 2$, let $\delta_i(x) = \text{dist}(x, \partial B_i)$ on B_i , extended by zero to $\mathcal{R}_a$. The interiors of the supports are disjoint. The unit L^p -sphere in $\text{span}\{\delta_1, \delta_2\}$ is odd-homeomorphic to S^1 , hence is an admissible compact symmetric set of genus two. Because the profiles are congruent and their supports are disjoint, every nonzero $c_1\delta_1 + c_2\delta_2$ has the same Rayleigh quotient. Polar integration gives

$$\int_{B_i} |\nabla \delta_i|^p = \pi r_a^2, \quad \int_{B_i} |\delta_i|^p = 2\pi \int_0^{r_a} (r_a - s)^p s \, ds = \frac{2\pi r_a^{p+2}}{(p+1)(p+2)}.$$

Therefore

$$\lambda_2(p; \mathcal{R}_a) \leq \frac{(p+1)(p+2)}{2r_a^p}. \tag{2}$$

For $u \in W_0^{1,p}(\mathcal{D}_a)$, apply sharp one-dimensional Dirichlet Poincaré inequality to almost every horizontal slice, then integrate in y . Since $|\nabla u|^p \geq |u_x|^p$,

$$\lambda_{\text{bar}}(p; \mathcal{R}_a) = \lambda_1(p; \mathcal{D}_a) \geq (p-1) \left(\frac{2\pi_p}{a} \right)^p \geq (p-1) \left(\frac{4}{a} \right)^p, \tag{3}$$

where

$$\pi_p = 2 \int_0^1 (1 - s^p)^{-1/p} \, ds \geq 2.$$

Condition (1) says exactly that the right-hand side of (2) is strictly smaller than the last expression in (3). This proves the comparison. Finally, $\rho_a < 1$ follows from $r_a > a/4$; its exponential decay dominates the polynomial prefactor in (1), so the condition holds eventually. $\square$

Corollary 1 (Finite last contact). *For every $1 < a < 2$, the set*

$$E_a = \{p \geq 2 : \lambda_2(p; \mathcal{R}_a) = \lambda_{\text{bar}}(p; \mathcal{R}_a)\}$$

is nonempty, closed, and bounded. Hence $p_{\text{last}}(a) = \max E_a < \infty$, and

$$\lambda_2(p; \mathcal{R}_a) < \lambda_{\text{bar}}(p; \mathcal{R}_a) \quad (p > p_{\text{last}}(a)).$$

Proof. The vertical two-half-rectangle test gives $\lambda_2(p; \mathcal{R}_a) \leq \lambda_{\text{bar}}(p; \mathcal{R}_a)$ for all $p > 1$. At $p = 2$, since $1 < a < 2$, the $(2, 1)$ rectangle mode is the second Laplace mode, and its eigenvalue is $\lambda_{\text{bar}}(2; \mathcal{R}_a)$. Thus $2 \in E_a$. Both branches are continuous in p , so E_a is closed. Theorem 1 makes it bounded. Its supremum is therefore a maximum, and after that maximum the only possible relation is strict inequality. $\square$

4 Scope, verification, and novelty

The theorem proves eventual strictness, and Corollary 1 proves the whole post-last-contact assertion, for every fixed $1 < a < 2$. It does not prove the conjectured equality throughout $[2, p_{\text{last}}(a)]$: that appears to require nonlinear branch uniqueness or a global optimal two-partition classification. At $a = 2$, the two-disk radius becomes $1/2 = a/4$, so the exponential gap vanishes. The anticipated $p^*(a) = +\infty$ for $a \geq 2$ therefore remains untouched.

The reproducible checker performs 29,997 dense-grid geometry checks and 14 finite-threshold checks. For $a = 1.01, 1.10, 1.25, 1.50, 1.75, 1.90, 1.99$, the first integer p certified by (1) is respectively 15, 23, 49, 195, 1375, 13016, 2209774. These computations audit the constants but do not replace the proof.

Four cheap run indexes and bounded exact-title, exact-conjecture, rectangle-symmetry, and p -to-infinity searches found no proof or later resolution through 11 August 2026. The elementary nature of the argument leaves an unindexed-folklore risk, so novelty confidence is moderate. Expert review should focus on the genus-two test family, the quotient calculation at the tangent supports, the one-dimensional eigenvalue normalization, and the continuity/last-contact inference.

Remark 3.4. In [20, Sections 4.2 and 4.3], HORÁK computed several lower energy eigenvalues of the p -Laplacian in the square and rectangles and formulated the following conjecture related to our analysis: For any $a > 1$, there exists $p^* = p^*(a) > 2$ such that

$$\begin{aligned} \lambda_2(p; \mathcal{R}_a) &= \lambda_{\square}(p; \mathcal{R}_a) \quad \text{for any } p \in [2, p^*], \\ \lambda_2(p; \mathcal{R}_a) &< \lambda_{\square}(p; \mathcal{R}_a) \quad \text{for any } p > p^*. \end{aligned}$$

Adding to this, we anticipate $p^* = +\infty$ for any $a \geq 2$.

Figure 1: Source PDF page 11: Remark 3.4 states Horák’s rectangle conjecture and the anticipated $a \geq 2$ regime.

References

- [1] V. Bobkov, *Non-Ljusternik-Schnirelman eigenvalues of the pure p -Laplacian exist*, arXiv:2604.01138 (2026).
- [2] J. Horák, *Numerical investigation of the smallest eigenvalues of the p -Laplace operator on planar domains*, Electronic Journal of Differential Equations 2011, no. 132, 1–30; arXiv:1106.0602.